

Hw Answers Chapter 3

2a. Java class definition header

<header> -> class <id> [extends <id>] [implements <id> { , <id> }]

2b. Java method

<call> -> <id> ([<exp_list>]) ;

<exp_list> -> <exp> { , <exp> }

2c. A C switch statement

<switch> -> switch (<exp>) { { <case> } [<default>] }

<case> -> case <const> : { <stmt> }

<default> -> default : { <stmt> }

2d. A C union

<union_def> -> union <id> { { <member> } } ;

<member> -> <type> <id> ;

2e. C Float Literals

<float> -> <int> . <int>

<int> -> <digit> { <digit> }

<digit> -> 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9

5.

<bexp> -> <bexp> || <bterm>
| <bterm>

<bterm> -> <bterm> && <bfactor>
| <bfactor>

<bfactor> -> ! <bfactor>
| (<bexp>)
| <rexp>
| true
| false

<rexp> -> <exp> <relop> <exp>

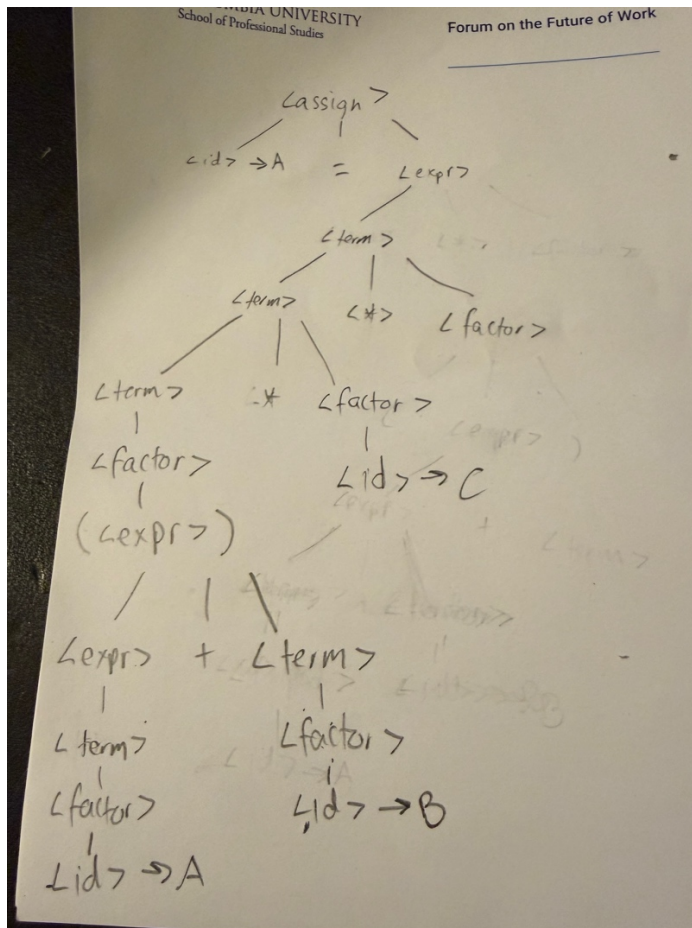
<relop> -> < | <= | > | >= | == | !=

<exp> -> <id> | <num>

7a) $A = (A + B) * C$

Leftmost derivation

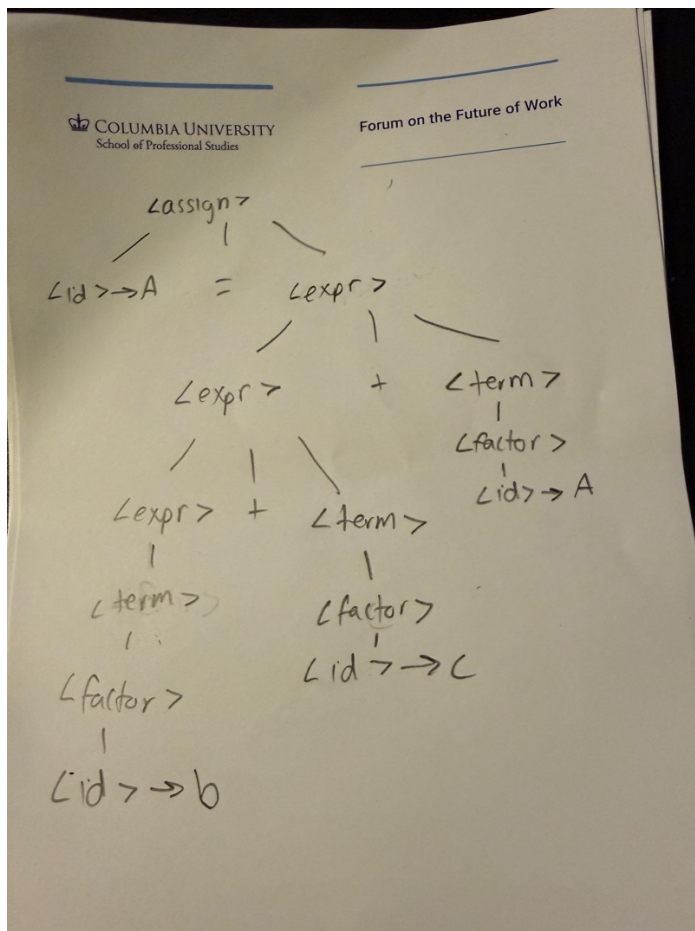
- $\langle \text{assign} \rangle$
- $\langle \text{id} \rangle = \langle \text{expr} \rangle$
- $A = \langle \text{expr} \rangle$
- $A = \langle \text{term} \rangle$
- $A = \langle \text{term} \rangle * \langle \text{factor} \rangle$
- $A = \langle \text{factor} \rangle * \langle \text{factor} \rangle$
- $A = (\langle \text{expr} \rangle) * \langle \text{factor} \rangle$
- $A = (\langle \text{expr} \rangle + \langle \text{term} \rangle) * \langle \text{factor} \rangle$
- $A = (\langle \text{term} \rangle + \langle \text{term} \rangle) * \langle \text{factor} \rangle$
- $A = (\langle \text{factor} \rangle + \langle \text{term} \rangle) * \langle \text{factor} \rangle$
- $A = (\langle \text{id} \rangle + \langle \text{term} \rangle) * \langle \text{factor} \rangle$
- $A = (A + \langle \text{term} \rangle) * \langle \text{factor} \rangle$
- $A = (A + \langle \text{factor} \rangle) * \langle \text{factor} \rangle$
- $A = (A + \langle \text{id} \rangle) * \langle \text{factor} \rangle$
- $A = (A + B) * \langle \text{factor} \rangle$
- $A = (A + B) * \langle \text{id} \rangle$
- $A = (A + B) * C$



7b $A = B + C + A$

Leftmost derivation

- $\langle \text{assign} \rangle$
- $\langle \text{id} \rangle = \langle \text{expr} \rangle$
- $A = \langle \text{expr} \rangle$
- $A = \langle \text{expr} \rangle + \langle \text{term} \rangle$
- $A = \langle \text{expr} \rangle + \langle \text{term} \rangle + \langle \text{term} \rangle$
- $A = \langle \text{term} \rangle + \langle \text{term} \rangle + \langle \text{term} \rangle$
- $A = \langle \text{factor} \rangle + \langle \text{term} \rangle + \langle \text{term} \rangle$
- $A = \langle \text{id} \rangle + \langle \text{term} \rangle + \langle \text{term} \rangle$
- $A = B + \langle \text{term} \rangle + \langle \text{term} \rangle$
- $A = B + \langle \text{factor} \rangle + \langle \text{term} \rangle$
- $A = B + \langle \text{id} \rangle + \langle \text{term} \rangle$
- $A = B + C + \langle \text{term} \rangle$
- $A = B + C + \langle \text{factor} \rangle$
- $A = B + C + \langle \text{id} \rangle$
- $A = B + C + A$



7c. $A = A * (B + C)$

$\langle \text{assign} \rangle$

$\langle \text{id} \rangle = \langle \text{expr} \rangle$

$A = \langle \text{expr} \rangle$

$A = \langle \text{term} \rangle$

$A = \langle \text{term} \rangle * \langle \text{factor} \rangle$

$A = \langle \text{factor} \rangle * \langle \text{factor} \rangle$

$A = \langle \text{id} \rangle * \langle \text{factor} \rangle$

$A = A * \langle \text{factor} \rangle$

$A = A * (\langle \text{expr} \rangle)$

$A = A * (\langle \text{expr} \rangle + \langle \text{term} \rangle)$

$A = A * (\langle \text{term} \rangle + \langle \text{term} \rangle)$

$A = A * (\langle \text{factor} \rangle + \langle \text{term} \rangle)$

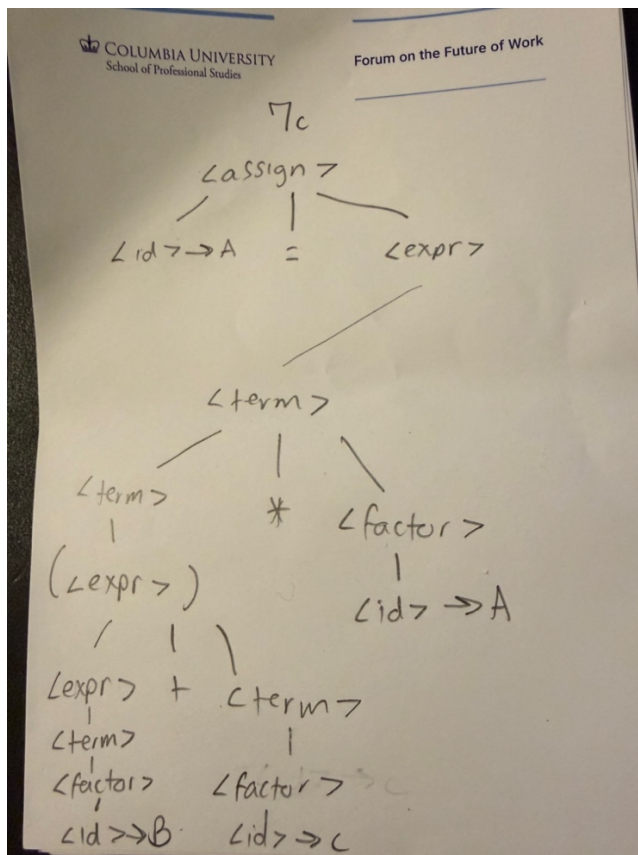
$A = A * (\langle \text{id} \rangle + \langle \text{term} \rangle)$

$A = A * (B + \langle \text{term} \rangle)$

$A = A * (B + \langle \text{factor} \rangle)$

$A = A * (B + \langle \text{id} \rangle)$

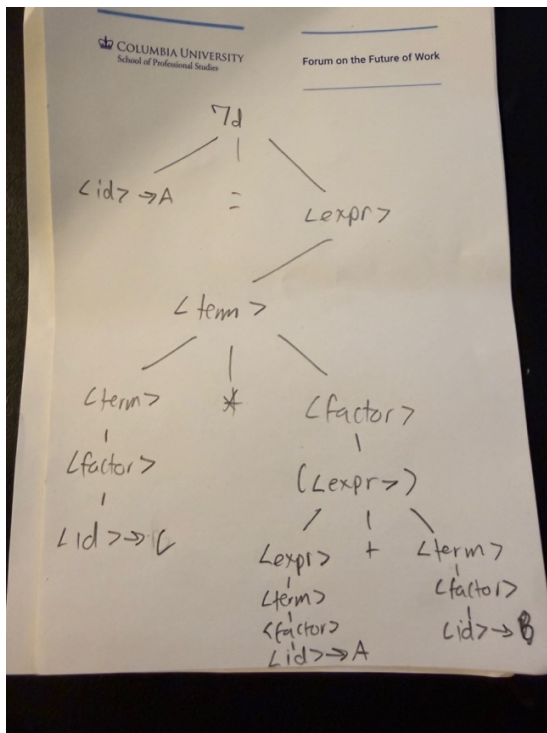
$A = A * (B + C)$



7d. $A = B * (C * (A + B))$

Leftmost derivation

- $\langle \text{assign} \rangle$
- $\langle \text{id} \rangle = \langle \text{expr} \rangle$
- $A = \langle \text{expr} \rangle$
- $A = \langle \text{term} \rangle$
- $A = \langle \text{term} \rangle * \langle \text{factor} \rangle$
- $A = \langle \text{factor} \rangle * \langle \text{factor} \rangle$
- $A = \langle \text{id} \rangle * \langle \text{factor} \rangle$
- $A = B * \langle \text{factor} \rangle$
- $A = B * (\langle \text{expr} \rangle)$
- $A = B * (\langle \text{term} \rangle)$
- $A = B * (\langle \text{term} \rangle * \langle \text{factor} \rangle)$
- $A = B * (\langle \text{factor} \rangle * \langle \text{factor} \rangle)$
- $A = B * (\langle \text{id} \rangle * \langle \text{factor} \rangle)$
- $A = B * (C * \langle \text{factor} \rangle)$
- $A = B * (C * (\langle \text{expr} \rangle))$
- $A = B * (C * (\langle \text{expr} \rangle + \langle \text{term} \rangle))$
- $A = B * (C * (\langle \text{term} \rangle + \langle \text{term} \rangle))$
- $A = B * (C * (\langle \text{factor} \rangle + \langle \text{term} \rangle))$
- $A = B * (C * (\langle \text{id} \rangle + \langle \text{term} \rangle))$
- $A = B * (C * (A + \langle \text{term} \rangle))$
- $A = B * (C * (A + \langle \text{factor} \rangle))$
- $A = B * (C * (A + \langle \text{id} \rangle))$
- $A = B * (C * (A + B))$



11. Answers for this are A and D

- It is not B because it is only one a ($k=1$)

It is not C because none contain s

21a. Java do { S } while (B);

```
loop:
  S
  if B != 0 goto loop
```

21b. Ada for I in L .. U loop S end loop;

```
I = L
loop:
  if I > U goto out
  S
  I = I + 1
  goto loop
out:
```

21c. C++ if (B) S1 else S2

```
if B == 0 goto else_part
  S1
  goto end_if
else_part:
  S2
end_if:
```

21d. C for (expr1; expr2; expr3) S

```
expr1
loop:
  if expr2 == 0 goto out
  S
  expr3
  goto loop
out:
```

24(a)

Statements:

1. $a = 2*b + 1$;
 2. $b = a - 3$;
- Postcondition: $\{ b < 0 \}$

Work backwards.

Step 1: WP of $b = a - 3$ w.r.t. $(b < 0)$

Replace b with $(a - 3)$:

- $(a - 3) < 0$
- $a < 3$

So: $WP(b = a - 3, b < 0) = (a < 3)$

Step 2: WP of $a = 2*b + 1$ w.r.t. $(a < 3)$

Replace a with $(2*b + 1)$:

- $2*b + 1 < 3$
- $2*b < 2$
- $b < 1$

Answer 24(a): $\{ b < 1 \}$

24(b)

Statements:

1. $a = 3 * (2*b + a)$;
 2. $b = 2*a - 1$
- Postcondition: $\{ b > 5 \}$

Work backwards.

Step 1: WP of $b = 2*a - 1$ w.r.t. $(b > 5)$

Replace b with $(2*a - 1)$:

- $2*a - 1 > 5$
- $2*a > 6$
- $a > 3$

So: $WP(b = 2a - 1, b > 5) = (a > 3)$

Step 2: WP of $a = 3*(2*b + a)$ w.r.t. $(a > 3)$

Replace a with $3*(2*b + a)$

$3*(2*b + a) > 3$

Divide by 3 (positive):

- $2*b + a > 1$

Answer 24(b): $\{ 2*b + a > 1 \}$